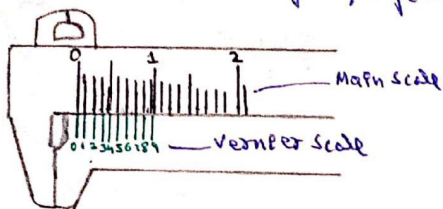


Vernier Caliper :

By using vernier caliper, we can measure

- 1) Inner diameter of any object
- 2) Depth of any object
- 3) Dimensions (L x B) of any object.



* A vernier caliper has two types of scale

- i.e.
- 1) Main scale
 - 2) Vernier scale

* In a standard vernier caliper with no error, 10 VSD coincide with 9 MSD

* The main scale is a cm-scale. So by its help, we can find length of 1 VSD as follows.

$$10 \text{ MSD} = 1 \text{ cm}$$

$$1 \text{ MSD} = \frac{1}{10} \text{ cm} \Rightarrow 1 \text{ MSD} = 0.1 \text{ cm}$$

Let, length of 1 VSD be x

$$\therefore 10 \text{ VSD} = 9 \text{ MSD}$$

$$10 \times x = 9 \times 0.1$$

$$\Rightarrow x = 0.09 \text{ cm} \quad (1 \text{ VSD})$$

Least Count of Standard Vernier Caliper $\Rightarrow$

When no. of VSD coinciding with no. of MSD is more ($\therefore 10 \text{ VSD}$ coincide with 9 MSD), then only we can apply

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$\Rightarrow LC = 0.1 \text{ cm} - 0.09 \text{ cm}$$

$$\Rightarrow LC = 0.01 \text{ cm}$$

$\Rightarrow$ If no. of VSD coinciding < no. of MSD, then this formula & value of LC is not applicable.

Taking Reading from Vernier Caliper $\Rightarrow$

Step-01: Vernier scale ka zero, main scale jin do readings ke beech ho, usme se left wali reading lo.

Step-02: Ab dekho ki Vernier scale ka konse number ka division MSD ke saath exactly coincide kar raha hai (n)

Step-03: Phir $(n \times LC)$ ko Main scale wali step 1

Experimental Physics

ke reading mein add kardo.

Ques While measuring length of box by vernier scale it was found that zero on vernier scale lie b/w 8.6 cm & 8.7 cm and 6th VSD coincide with MSD, then what is its reading?

$$\begin{aligned} \text{Sol}^n \quad \text{Reading} &= 8.6 + (n \times LC) \\ &= 8.6 + (6 \times 0.01) \Rightarrow 8.6 + 0.06 \\ &= 8.66 \text{ cm} \quad \text{Ans} \end{aligned}$$

Error in Vernier Caliper $\Rightarrow$

Positive Zero Error

* When vernier scale zero lies ahead of Main scale zero.



* It is subtracted from final reading

$$\text{Reading} = \text{MSR} + (n \times LC) - E$$

Negative Zero Error

* When vernier scale zero lies behind the main scale zero.



* It is added to the final reading.

$$\text{Reading} = \text{MSR} + (n \times LC) + E$$

$\rightarrow$ To calculate the zero error, we simply see that which VSD coincide with the MSD and then multiply it by LC.

$\rightarrow$ In above both examples, 5th VSD coincide with the MSD, so,

$$\begin{aligned} \text{Error} &= 5 \times 0.01 \\ &= 0.05 \text{ cm} \end{aligned}$$

When more no. of MSD coincide with less no. of VSD $\Rightarrow$

Ques In a vernier caliper, if 10 VSD coincide with 11 MSD and during measurement, the zero of vernier scale is noted b/w 2.8 cm & 2.9 cm along with 0.4 of negative zero error, what will be the final reading?

$$\text{Sol}^n \quad \therefore 10 \text{ MSD} = 1 \text{ cm} \Rightarrow 1 \text{ MSD} = 0.1 \text{ cm}$$

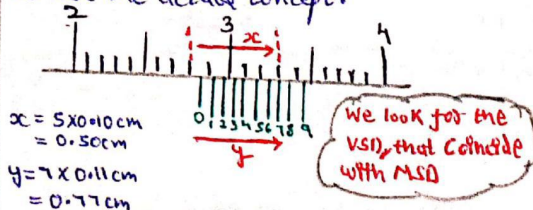
Now,

$$10 \text{ VSD} = 11 \text{ MSD}$$

$$10 \times 1 \text{ VSD} = 11 \times 0.1$$

$$1 \text{ VSD} = 0.11 \text{ cm}$$

Here, we can't use $LC = 1 \text{ MSD} - 1 \text{ VSD}$, thus we use the actual concept.



We look for the VSD, that coincide with MSD

$$x = 5 \times 0.10 \text{ cm} = 0.50 \text{ cm}$$

$$y = 7 \times 0.11 \text{ cm} = 0.77 \text{ cm}$$

$$\text{Reading} = 2.8 + (y - x) + E$$

$$= 2.8 + 0.22 + 0.4 = 3.42 \text{ cm} \quad \text{Ans}$$

Ques The diameter of a cylinder is measured using a vernier caliper with no zero error. It is found that zero of vernier scale lie between 5.1 cm & 5.2 cm of main scale. The vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of VS exactly coincide with one of MSD. The diameter of cylinder is

$$\begin{aligned} \text{Sol}^n \quad 1 \text{ MSD} &= \text{Diff. b/w two successive divisions} \\ &= 5.15 \text{ cm} - 5.10 \text{ cm} = 0.05 \text{ cm} \end{aligned}$$

$$1 \text{ VSD} = \frac{2.45}{50} = 0.049 \text{ cm}$$

$$\begin{aligned} \Rightarrow \text{Least Count (LC)} &= 1 \text{ MSD} - 1 \text{ VSD} \\ &= 0.050 - 0.049 \\ \text{LC} &= 0.001 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Reading} &= 5.10 + (24 \times 0.001) \\ &= 5.124 \text{ cm} \quad \text{Ans} \end{aligned}$$

Screw Gauge :

It is used to measure diameter of thin wires or size of objects.

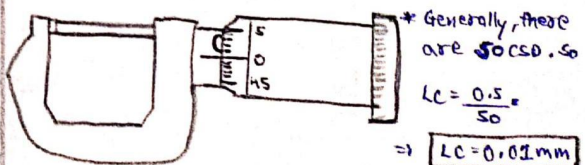
$\rightarrow$ It also have two scales $\rightarrow$ Main scale $\rightarrow$ Circular scale.

$\rightarrow$ Usually, $1 \text{ MSD} = 0.5 \text{ mm}$

* Pitch $\Rightarrow$ Pitch is the length of measurement opened up on Main scale by one complete revolution of the circular part in CW sense.

$$\text{Also, } \text{Least Count (LC)} = \frac{\text{Pitch}}{n} \quad (n = \text{No. of divisions on circular scale})$$

$\rightarrow$ In an 'error-free' screw gauge, the Main scale zero is partially visible.



* On Rotating Circular Part, Band hoto
(1) Anti-clockwise $\Rightarrow$ Moves left side (Khatta hai)
(2) Clockwise $\Rightarrow$ Moves right side (Khutta hai)

Errors in Screw Gauge $\Rightarrow$

+ve zero Error



Main scale zero completely visible at rest position

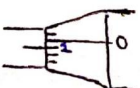
-ve zero Error



Main scale zero completely hidden at rest position

* $MSR = (LC \times n) - E$ * $MSR = (LC \times n) + E$

* $Error = LC \times (\text{zero se konse no. ka CSP, MSD se coincide kar raha hai, woh})$

Eg:  $Error = 0.01 \times 3 = 0.03 \text{ mm}$
 Since main scale zero is hidden, it is -ve zero error.

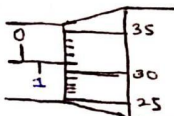
How to take Reading?

Step-01: Yeh dekho ke main scale pe left side konse no. ka MSD hai (1 MSD = 0.5 mm)

Step-02: Ab dekho ke konse number ka CSP, MSD ke zero se coincide kar raha hai.

$CSR = (n \times LC)$

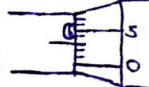
Step-03: $MSR = (n \times LC) + (\text{Main scale} \times 0.5 \text{ division})$

Eg:  $MSR = 0.5 + (31 \times 0.01) = 0.5 + 0.31$
 $MSR = 0.81 \text{ mm}$

Ques The pitch and no. of divisions of circular scale for a screw gauge are 0.5 mm and 100. When screw gauge is fully tightened w/o any object, the zero of its circular scale lies 3 divisions below the mean line.

The reading of main scale and circular scale for a thin sheet are 5.5 mm & 48. The thickness of this sheet is ____.

Sol: $LC = \frac{0.5}{100} = 0.005 \text{ mm}$ [JEE MAIN 2014]

 $Error = LC \times 3 = 0.005 \times 3$
 $E = 0.015 \text{ mm}$
 +ve zero error

Reading = $5.5 + (0.005 \times 48) - 0.015$
 $= 5.5 + 0.240 - 0.015$
 $= 5.5 + 0.225$

Reading = 5.725 mm Ans

Ques A screw gauge with a pitch 0.5 mm and circular scale with 50 div is used to measure thickness of a thin aluminium sheet. Before starting measurement, it was found that when two jaws of screw gauge are brought together, the 45th CSP coincides with mean line and zero of main scale is barely visible. Find thickness of sheet if MSR is 0.5 mm and 25th CSP coincides with MSD.

[JEE MAIN 2016]

Sol: $E = 5 \times 0.01 \Rightarrow 0.05 \text{ mm}$ (-ve error bcz zero barely visible)

Reading = $0.5 + (0.01 \times 25) + 0.05$
 $= 0.5 + 0.30 \Rightarrow 0.8 \text{ mm}$ Ans

Travelling Microscope:

→ Used to measure refractive index of a glass slab.

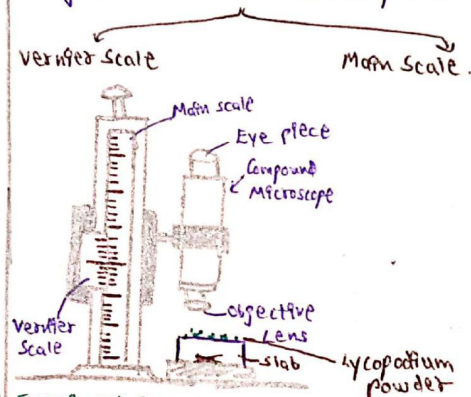
→ Works on the principle that,

$Refractive\ index\ (\mu) = \frac{\text{Real depth}}{\text{Apparent depth}}$

→ In travelling microscope, a compound microscope is fixed on vertical slide.

→ It has two scales → Vertical scale
 → Horizontal scale

→ Only vertical scale is of our use, which is further divided into two parts.



Experiment steps:

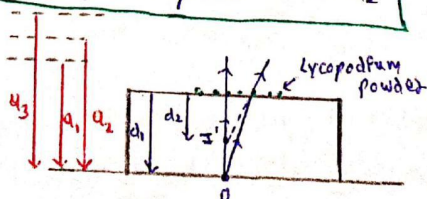
Step-01: First, a paper with a cross is observed from microscope such that cross is seen clearly and reading (a_1) is noted.

Step-02: Then, glass slab is placed over the paper and cross is observed clearly by moving microscope up and reading (a_2) is noted.

Step-03: Now Lycopodium powder/chalk powder is put on upper surface of slab.

Step-04: The microscope is further moved up to observe the powder clearly this time, and reading (a_3) is noted.

$\mu = \frac{\text{Real depth}(d_1)}{\text{Apparent depth}(d_2)} = \frac{a_3 - a_1}{a_3 - a_2}$



Ques In an experiment of measuring RI of a glass slab using travelling microscope, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of slab as 5.00 mm. Travelling microscope has 20 div. in one cm on main scale and 50 div. on vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is ____.

[JEE MAIN 2023]

Sol: It is given,

$d_1 = 5.25 \text{ mm}$ $\mu = \frac{d_1}{d_2} = \frac{5.25}{5.00} = \frac{21}{20}$
 $d_2 = 5.00 \text{ mm}$

Now, for the scale

$20 \text{ MSD} = 1 \text{ cm} \Rightarrow 1 \text{ MSD} = \frac{1}{20} \text{ cm}$

Also,

$50 \text{ VSD} = 49 \text{ MSD} \Rightarrow 1 \text{ VSD} = \frac{49 \text{ MSD}}{50}$

So,

Least Count (LC) = $1 \text{ MSD} - 1 \text{ VSD}$

$= 1 \text{ MSD} - \frac{49 \text{ MSD}}{50} = \frac{1}{50} \text{ MSD}$

$LC = \frac{1}{50} \times \frac{1}{20} \text{ cm} = 0.01 \text{ mm}$

Hence, the uncertainty (error) can be given by

$\mu = \frac{d_1}{d_2}$ (These will be LC of scale)

$\Rightarrow \frac{\Delta \mu}{\mu} = \frac{\Delta d_1}{d_1} + \frac{\Delta d_2}{d_2}$

$\frac{\Delta \mu}{\frac{21}{20}} = \frac{0.01}{5.25} + \frac{0.01}{5.00}$

$\Delta \mu \times \frac{20}{21} = \frac{1}{25} \left(\frac{21+20}{21 \times 20} \right) \Rightarrow \frac{41}{500}$

$\Delta \mu = \frac{41}{10000} \Rightarrow \Delta \mu = \frac{41}{10} \times 10^{-3}$

So,

$x = 41$ Ans

→ Measurement of a physical quantity contains two components.

$$Q = nU \quad \text{Unit}$$

Numerical Value

Also,

$$n_1 u_1 = n_2 u_2$$

Important Quantities & their dimensions :

Physical Quantity	Dimensions
Force	MLT^{-2}
Energy	$ML^2 T^{-2}$
ϵ_0	$M^{-1} L^{-3} T^4 A^2$
μ_0	$ML T^{-2} A^{-2}$

Principle of Homogeneity :

In any uniform physical expression/ equation, dimensions of all quantities involved is always same.

i.e. $a = b + c$

$$\Rightarrow [a] = [b] + [c]$$

Ques Expression for time in terms of gravitational constant (G), Planck's constant (h) and speed of light (c) is proportional to?

Sol: $t \propto G^a h^b c^c$

$$[t] = [G]^a [h]^b [c]^c$$

$$(M^0 L^0 T^1) = (M^{-1} L^3 T^{-2})^a (ML^2 T^{-1})^b (LT^{-1})^c$$

$$M^0 L^0 T^1 = M^{-a+b} L^{3a+2b+c} T^{-2a-b-c}$$

$$\Rightarrow a=b=1/2 \quad \text{and} \quad c=-5/2$$

$$So, \quad t \propto \sqrt{\frac{Gh}{c^5}} \quad \text{Ans}$$

Dimensionless Quantities :

1) All trigonometric ratios ($\sin \theta$, $\cos \theta$, $\tan \theta$)

2) Angle θ (If $\sin(\frac{ab}{c}) \Rightarrow [ab] = [c]$)

3) Exponential function,

$e^x \rightarrow$ Must be dimensionless.

i.e. $(e^{-t/RC} \Rightarrow [t] = [RC])$

4) Reynold's No. ($Re = \frac{\rho v D}{\eta}$), Dielectric constant (K) and refractive index.

Units, Dimensions & Errors

Error Analysis :

$$(\text{Error} = \text{True value} - \text{Measured value})$$

Step-01: We always take R_{avg} as true value

$$R_{avg} = \frac{R_1 + R_2 + \dots + R_n}{n}$$

Step-02: Absolute Error $\Rightarrow$

$$\Delta R_1 = R_1 - R_{avg}$$

$$\Delta R_2 = R_2 - R_{avg}$$

$$\vdots$$

$$\Delta R_n = R_n - R_{avg}$$

[Take only magnitudes of ΔR_i]

Step-03: Mean absolute Error $\Rightarrow$

$$\Delta R_{avg} = \frac{|\Delta R_1| + |\Delta R_2| + \dots + |\Delta R_n|}{n}$$

$$\text{Final Reading} = R_{avg} \pm \Delta R_{avg}$$

Real Error

$$\frac{\Delta R_{avg}}{R_{avg}}$$

% Error

$$\frac{\Delta R_{avg}}{R_{avg}} \times 100\%$$

Combination of Error :

Eg: Error is Δx , Δy & Δz . Find error in P , if $P = d \frac{x^a y^b}{z^c}$.

Sol:

$$\frac{\Delta P}{P} = a \cdot \frac{\Delta x}{x} + b \cdot \frac{\Delta y}{y} + c \cdot \frac{\Delta z}{z}$$

* In case of instruments like vernier calliper, screw gauge and travelling microscope,

$$\Delta x = \text{Least Count.}$$

Important Unit Conversions :

1) 1 sec of arc = $\frac{1^\circ}{3600} = \frac{1}{3600} \times \frac{\pi}{180} \text{ rad}$

2) 1 min. of arc = $\frac{1^\circ}{60} = \frac{1}{60} \times \frac{\pi}{180} \text{ rad.}$

3) 1 day = 86400 sec

4) 1 dyne = 10^{-5} N

5) 1 erg = 10^{-7} J

6) 1 calorie = $4.186 \text{ J} \approx 4.2 \text{ J}$

7) 1 kWh = $10^3 (\text{J/s}) \times 3600(\text{s}) = 3.6 \times 10^6 \text{ J}$

8) 1 eV = $1.6 \times 10^{-19} \text{ J}$

(SI unit: $\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$)

9) 1 Poiseuille = 10 Poise (Viscosity units)

10) For volume

$$1 \text{ Ltr} = 10^{-3} \text{ m}^3$$

$$1 \text{ cm}^3 = 10^{-6} \text{ m}^3$$

11) 1 MPa = 10^6 Pa

$$1 \text{ GPa} = 10^9 \text{ Pa}$$

12) 1 rpm = $\left(\frac{2\pi}{60}\right) \text{ rad/s}$

$$1 \text{ rps} = 2\pi \text{ rad/s}$$